Revised Dissertation Plan

Interactive Proofs for Resource-Constrained Quantum Learning

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April 2026 — Revision 2

1 Overview

This document is the revised plan for the CS310 final report. It incorporates three inputs: (i) Dr. Caro's feedback on the original plan (March 2026 meeting and written comments), (ii) the format requirements at https://warwick.ac.uk/fac/sci/dcs/teaching/material/cs310/components/final/format/, and (iii) the completed experiment findings (11 experiments, April 2026).

Core thesis (revised). The MoS verification protocol for agnostic parity learning can be faithfully implemented and its completeness, soundness, and robustness properties empirically characterised across the full parameter space—including adversarial provers, label-flip and gate-level noise, Fourier-dense targets, and system sizes $n = 4$ to 16. The empirical results precisely confirm over ten theoretical predictions from Caro et al. (ITCS 2024), identify where theory–practice gaps arise (precondition violations, tight weight margins, circuit noise), and provide novel evidence on gate noise sensitivity that lies outside the paper's theoretical scope.

Target length: 8,000 words (max 10,000). Front/back matter excluded.

2 Chapter Plan

Word counts are approximate targets summing to ∼8,500.

Front Matter (excluded from count)

Title page (title, author, supervisor, year). Abstract ($\leq$ 300 words): research question, approach, key results. Keywords (6–8): quantum machine learning, interactive proofs, Fourier sampling, mixture of superpositions, classical verification, agnostic learning, Qiskit simulation. Table of Contents.

2.1 Chapter 1 — Introduction ∼1,000 *words*

1.1 Motivation and Context. QML promises computational advantages, but cloud delegation creates a trust problem: how does a classical client verify that an untrusted quantum server performed the correct learning task? Brief framing of why this matters as quantum hardware scales.

1.2 Research Question. Can a classical client efficiently verify quantum agnostic parity learning via the mixture-of-superpositions technique?

1.3 Objectives and Contributions. Revised to match what was actually achieved:

(i) Faithful implementation of the distributional MoS verification protocol (Theorems 12 and 15 of Caro et al.) in Qiskit, comprising prover, verifier, and MoS simulator modules.
(ii) Systematic empirical confirmation of 10+ theoretical predictions (postselection rate, Parseval bound, noise attenuation, completeness, soundness).
(iii) Identification of theory–practice gaps: precondition violations for large k, tight weight margins, the weight check as the sole operative rejection mechanism.
(iv) Novel empirical characterisation of gate-level depolarising noise, including a clean explanatory model for the $n = 5 \to 6$ phase transition ($1/2^n$ vs. $\theta^2/4$).

(v) Empirical investigation of verifier truncation, θ-sensitivity, bent-function worst cases, and the $[a^2, b^2]$ distributional regime.

1.4 Report Structure. Roadmap of chapters.

2.2 Chapter 2 — Background ~1,600 *words*

Restructured per Dr. Caro's feedback to unify three pillars (quantum, learning, Fourier) into a coherent arc. Each subsection ends with a forward pointer explaining how it connects to the next.

2.1 Quantum Computation Preliminaries (~250 words). Qubits, measurement, Hadamard transform. Only the material needed to understand the protocol circuits. *Forward pointer:* the Hadamard transform connects quantum measurement to Fourier coefficients (developed in §2.5).

2.2 Verifying Quantum Computation (~350 words). The QPIP model: classical client, quantum server, classical messages. Completeness and soundness (Definition 7). One–two sentences on oracle separations ($\mathrm{BQP}^A \not\subset \mathrm{MA}^A$) as motivation for why verification is non-trivial. Mahadev's protocol: state the result (LWE-based, computational soundness), describe the high-level structure (TCF commitment, Hadamard round) in ≤ 250 words. Do *not* critique the mixed-state obstacle here—defer to Discussion (§6.4). *Forward pointer:* Mahadev verifies arbitrary quantum computation; can we do better when the computation has learning-theoretic structure?

2.3 PAC Learning and Agnostic Learning (~300 words). PAC model, agnostic PAC model, α-agnostic learner. Parity learning and LPN hardness. Define the benchmark class $\mathcal{B}$ (parity functions, k-sparse Fourier sums). *Forward pointer:* the connection between learning parities and Fourier analysis.

2.4 Fourier Analysis for Classical and Quantum Learning (~350 words). Boolean Fourier expansion, Parseval's identity, heavy Fourier coefficients. Quantum Fourier Sampling (QFS): superposition of labelled examples, Hadamard + measure $\rightarrow$ approximate sampling from $|\hat{\varphi}(s)|^2$. Theorem statement (Theorem 5). This is the bridge: Fourier structure makes learning *and* verification efficient. *Forward pointer:* the MoS construction resolves the distributional vs. functional gap that prevents direct QFS.

2.5 Verifying Quantum Learning (~250 words). Position Caro et al. (ITCS 2024) as combining all three pillars. Brief survey of related work beyond the immediate references (per Dr. Caro's feedback): Ma–Su–Deng (2024) on noise robustness; Caro et al. impossibility results (arXiv:2410.23969); Goldwasser–Kalai et al. on interactive proofs for delegated computation (the classical analogue); compiled nonlocal games. Note that parity is currently the only setting with an information-theoretic verification protocol.

Connecting diagram. A figure showing the three pillars (quantum computation, learning theory, Fourier analysis) converging on "verifying quantum learning," with Mahadev and Goldwasser–Kalai positioned as related but distinct approaches.

Budget discipline: every paragraph must serve a later chapter. The oracle separation material is at most 1–2 sentences. Mahadev is ≤ 250 words here.

2.3 Chapter 3 — The MoS Verification Protocol ~1,300 *words*

3.1 Mixture of Superpositions Construction (~250 words). Definition of F_D, the mixed state $\rho_D = \mathbb{E}_{f\sim F_D}[|\psi_f\rangle\langle\psi_f|]$. Why MoS resolves the distributional vs. functional gap.

3.2 Approximate Quantum Fourier Sampling from MoS (~200 words). $H^{\otimes(n+1)}$ + measure $\rightarrow$ postselect on $b = 1$. Output distribution approximates $|\hat{\varphi}(s)|^2$. Sample complexity $O(1/\vartheta)$.

3.3 The Three-Step Protocol (~350 words). Step 1: Prover extracts heavy coefficients L via QFS. Step 2: Verifier estimates correlations using classical examples. Step 3: Weight check and hypothesis construction. Information-theoretic soundness argument.

Include **pseudocode** (Algorithm 1) for the three-step protocol. Consider a second pseudocode block for the extraction procedure (Corollary 5). This communicates the structure and scale of the codebase without dumping code listings.

3.4 Protocol Parameters (~200 words). Relationship between ε, δ, ϑ, $|L|$, sample complexities. How these translate to experimental parameters.

3.5 Open Questions (~150 words). Strategically highlight what is missing (per Dr. Caro's feedback), now grounded in the experiment findings:

(a) No lower bounds on sample complexity—is $O(1/\vartheta)$ tight?
(b) The weight check (Step 4) is the sole operative rejection mechanism across all 11 experiments; the list-size bound (Step 3) never triggers. Is there *any* function class for which Step 3 is binding?
(c) No theoretical treatment of circuit noise. The gate noise experiment (§5.4) provides the first empirical evidence.
(d) Corollary 5's $\varepsilon > 2^{-(n/2-2)}$ lower bound appears non-tight: the $n = 4$ bent-function results work despite violating it.
(e) Remark 4 (prover sends coefficient estimates, verifier uses $O(1)$ queries) is left unproven. [Deferred to Future Work, §7.2.]

Table 3.1: Protocol parameters and their theoretical relationships.
Figure 3.1: MoS construction diagram.
Figure 3.2: Three-step protocol flow (Prover → list L → Verifier → accept/reject + hypothesis).

2.4 Chapter 4 — Implementation ∼1,000 *words*

4.1 Architecture Overview (∼250 words). Module structure: `mos/` (simulator), `ql/` (prover + verifier). Brief statement of codebase scale (∼ X lines across Y modules). Design decisions: why Qiskit, why statevector simulation, why three execution modes (statevector, circuit, batched).

4.2 MoS Simulator (∼200 words). State preparation via oracle circuits. Per-shot f-sampling to realise the mixture. Noise injection: $\varphi_{\text{eff}}(x) = (1 - 2\eta)\varphi(x) + \eta$ (Definition 5(iii)).

4.3 Quantum Prover (∼200 words). Postselection on label qubit = 1. Corollary 5 extraction: empirical conditional distribution with threshold $\theta^2/4$. Discussion of the extraction threshold's interaction with system size (the $1/2^n$ vs. $\theta^2/4$ crossover observed in the gate noise experiment).

4.4 Classical Verifier (∼200 words). Classical example oracle. Fourier coefficient estimation via $\mathbb{E}[(1-2y)(-1)^{\langle s,x\rangle}]$. Weight interval check with tolerance. Hypothesis construction.

4.5 Validation (∼150 words). Comparison of empirical $\Pr[s \mid b = 1]$ against theoretical distribution at small n. Statevector vs. circuit mode agreement.

Figure 4.1: Module architecture diagram.
Figure 4.2: Example Qiskit circuit for $n = 3$ MoS measurement.
Figure 4.3: Empirical vs. theoretical $\Pr[s \mid b = 1]$ validation plot.

Note: "Noise floor estimation" and "density matrix purity check" removed from the plan (per Dr. Caro's feedback—explain clearly or cut).

2.5 Chapter 5 — Experiments and Results ∼2,100 *words*

Grouped thematically by protocol property, not listed sequentially. Each experiment opens with the specific theoretical claim it tests.

5.1 Experimental Setup (∼200 words)

Two parameter regimes (single-parity and multi-coefficient targets), function families, common parameters (100 trials per cell, $\varepsilon = 0.3$, $\delta = 0.1$), resource budget (6,000 samples per trial). Note that the implementation uses the *distributional* protocol (Theorems 12/15) throughout.

5.2 Completeness (∼450 words)

Exp 1.1 — Scaling baseline (Theorems 5, 12). Perfect completeness $n = 4$–16 for single parities. Postselection rate $= 0.50 \pm 0.003$, confirming Theorem 5(i). $|L| = 1$ throughout; wall-clock growth is a simulation artefact.

Exp 1.2 — Average-case performance (Corollary 7 / Theorem 15). $|L|$ tracks sparsity k, not dimension n. Completeness below $1 - \delta$ for non-trivial function families. Root causes: theorem precondition violations (coefficients below θ), tight weight margins, random booleans outside promise class.

Exp 1.3 — k-Sparse verification (Corollary 7 / Theorem 15). Misclassification bound holds for $k \leq 4$; violated at $k = 8$ due to precondition failures. Weight threshold is the binding constraint; anomalous

small-n acceptance explained by brute-force enumeration of 2^n strings.

5.3 Soundness (∼350 words)

Exp 2.1 — Single-parity dishonest provers (Theorem 12 soundness). Four adversarial strategies tested across $n = 4$–20. Three of four achieve 100% rejection; random-list rejection modelled by $\Pr[s^* \notin L] \approx 1 - 5/2^n$. All rejections via weight check (Step 4); list-size bound never triggers.

Exp 2.2 — Multi-element dishonest provers (Theorem 15 soundness). Four strategies at $k \in \{2, 4\}$, $n = 4$–16. Soundness maintained for all tested (strategy, k) combinations. All rejections via weight check.

5.4 Robustness (∼500 words)

Exp 3.1 — Label-flip noise sweep (Definition 5(iii), Lemma 6). Weight attenuation precisely tracks $(1 - 2\eta)^2$ with $< 3\%$ relative error. Non-monotonic acceptance pattern explained by relative margin growth. No breakdown up to $\eta = 0.40$; theoretical breakdown at $\eta_{\max} = 0.447$.

Exp 3.2 — Gate-level depolarising noise (no existing theorem). *Novel finding.* Sharp $n = 5 \to 6$ phase transition. Explanatory model: for $n \leq 5$, $1/2^n \geq \theta^2/4$, so even fully corrupted QFS includes the correct string by brute force. Crossover at $n = \log_2(4/\theta^2) = 5.47$ matches data exactly. Gate noise is qualitatively different from label-flip noise; no formal equivalence exists.

Exp 3.3 — $a^2 \neq b^2$ regime (Definition 14, Theorem 13). Wider gap lowers acceptance threshold; Theorem 13 worst-case bound ($\varepsilon \geq 2\sqrt{b^2 - a^2}$) is not tight for typical functions.

5.5 Sensitivity and Practical Limits (∼400 words)

Exp 4.1 — Bent functions (Corollary 1/5). Sharp phase transition between $n = 4$ (100% acceptance, $|L| = 16$) and $n = 6$ (0%, $|L| \sim 1$). QFS output is uniform for bent functions; Fourier sparsity is load-bearing.

Exp 4.2 — θ-sensitivity (Theorem 4/Corollary 5). Practical sweet spot at $\theta = 0.10$–0.15. Parseval bound $4/\theta^2$ universally satisfied. Postselection rate independent of θ, confirming Theorem 5(i).

Exp 4.3 — Verifier truncation (Theorem 12, Step 3 complexity). Counterintuitive pattern at tight margins: more samples → lower acceptance. Explanation: the verifier is becoming *more accurate*, correctly identifying that the margin is razor-thin. At wide margins ($\varepsilon = 0.5$), normal pattern holds.

5.6 Summary (∼200 words)

Theory-vs-empirics comparison table (adapted from the findings document, §5.1). Cross-cutting observations: weight check dominance, Fourier sparsity as load-bearing assumption, sources of theory–practice gaps.

Planned figures and tables:

- Table 5.1: Theory-vs-empirics verdict for each tested prediction.
- Fig. 5.1: Noise sweep—$(1 - 2\eta)^2$ attenuation fit and non-monotonic acceptance.
- Fig. 5.2: Gate noise—$n = 5 \to 6$ transition with $1/2^n$ vs. $\theta^2/4$ annotated.
- Fig. 5.3: Bent function phase transition ($|L|$ and acceptance vs. n).
- Fig. 5.4: θ-sensitivity—$|L|$ and acceptance vs. θ at $n = 16$.
- Fig. 5.5: Truncation—acceptance vs. verifier samples at multiple ε.
- Table 5.2: Dishonest prover rejection rates by strategy and n.
- Table 5.3: k-sparse misclassification vs. theoretical bound.

2.6 Chapter 6 — Discussion ∼1,100 *words*

6.1 Theory Confirmation and Where Bounds Are Tight (∼300 words). Postselection rate, Parseval bound, $(1 - 2\eta)^2$ attenuation: all precisely confirmed. Theorem 13 worst-case bound is not tight for typical functions. Corollary 5's $\varepsilon > 2^{-(n/2-2)}$ lower bound is non-tight ($n = 4$ bent functions work despite violating it).

6.2 Where the Protocol Breaks and Why (∼300 words). Weight check (Step 4) is the sole operative rejection mechanism across all experiments—the list-size bound never triggers. Precondition violations for large k: Dirichlet-drawn coefficients fall below θ, violating $\mathcal{D}^{\text{func}}_{U_n;\geq\theta}$. Bent-function phase transition: sharp, not gradual, driven by detection boundary. The gap between distributional and functional theorems.

Explicitly map results back to the five objectives in §1.3 (per Dr. Caro's feedback: "Have you answered the research question? Which objectives have you achieved?").

6.3 Gate Noise as an Open Frontier (∼200 words). The $1/2^n > \theta^2/4$ explanation for small-n acceptance. No formal equivalence to label-flip noise; at $p = 0.01$ the protocol fails completely for $n \geq 6$, while $\eta = 0.01$ (label-flip) shows no degradation. What a theoretical treatment would need: a noise model that acts on the QFS circuit itself (Hadamard layer + oracle), not just the data distribution.

6.4 Limitations (∼200 words). For each limitation, state how much it affects the conclusions:

- Statevector simulation: no true quantum resource accounting; wall-clock scaling is a simulation artefact.
- No Goldreich–Levin extraction: limits claims about practical scalability beyond direct thresholding. The protocol scales to $n = 16$ but with $|L| = 1$ for single parities, which does not stress-test the extraction.
- Weight interval computed from true Fourier spectrum, not a distributional promise.
- The Mahadev connection: direct composition fails because ρ_D is mixed across independently sampled f, breaking the spot-checking mechanism's pure-state requirement. This is a fundamental theoretical boundary, not an engineering gap. ($\leq$ 100 words; kept brief since Mahadev is already covered in §2.2.)

Per Dr. Caro: the discussion must not be lost. This chapter is the most important upgrade from the original plan.

2.7 Chapter 7 — Conclusions and Future Work ∼400 ***words***

7.1 Summary of Contributions (∼200 words). First paragraph directly answers the research question from §1.2. Not "we implemented X and ran Y experiments" but "a classical client can verify quantum parity learning via MoS in the following regimes, with the following caveats."

7.2 Future Work (∼200 words).

- Goldreich–Levin extraction at scale ($n \geq 16$) for multi-coefficient targets.
- Remark 4 protocol variant (prover sends coefficient estimates, $O(1)$ verifier queries)—theoretical development.
- Real quantum hardware execution.
- Ma–Su–Deng error rectification implementation.
- Theoretical treatment of gate-level noise (formalising the $1/2^n$ vs. $\theta^2/4$ crossover).
- Non-parity concept classes; non-uniform marginals.

Back Matter (excluded from count)

References. Aharonov et al. 2017, Mahadev 2018, Regev 2005/2024, Caro et al. ITCS 2024, Caro et al. 2024 (impossibility), Ma–Su–Deng 2024, Goldwasser–Kalai et al., and implementation references (Qiskit, etc.).

Appendices (if needed). Pseudocode for extraction procedure (if not in Chapter 3). Extended experimental data tables (per-cell results for all 11 experiments). Selected code listings with inline comments explaining design decisions (e.g., why $\theta^2/4$ as extraction threshold, why bottom-quartile noise floor heuristic). No substantial code dumps.

Author's Self-Assessment. ∼100-word answers to the five self-assessment questions from the format specification. The "technical contribution" and "why an achievement" answers connect to §6.2 (research question answered) and §6.3 (novel gate noise finding).

Chapter	Original	Revised	Rationale
1. Introduction	1,000	1,000	Rewrite objectives to match actual contributions
2. Background	1,800	1,600	Restructured per Caro; cut oracle separations, tighten Mahadev
3. Protocol	1,200	1,300	Add pseudocode, open questions grounded in findings
4. Implementation	1,200	1,000	Cut jargon; interesting content is in experiments
5. Experiments	2,000	2,100	11 experiments grouped thematically
6. Discussion	800	1,100	Most important upgrade; concrete content from findings
7. Conclusions	300	400	Direct answer to research question
Total	**8,300**	**∼8,500**	Within 8,000–10,000 range

3 Word Budget

4 Planned Figures and Tables

Chapter 2 — Background

- Fig. 2.1: Three-pillar connecting diagram (quantum computation, learning theory, Fourier analysis → verifying quantum learning). Positions Mahadev and Goldwasser–Kalai as related approaches.
- Fig. 2.2: QPIP client–server model (classical messages, quantum prover).

Chapter 3 — Protocol

- Fig. 3.1: MoS construction—sampling $f \sim F_D$, preparing $|\psi_f\rangle$, averaging to ρ_D.
- Fig. 3.2: Three-step protocol flow diagram.
- Table 3.1: Protocol parameters and theoretical relationships (ε, δ, ϑ, $|L|$, sample complexities).
- Algorithm 1: Three-step protocol pseudocode.

Chapter 4 — Implementation

- Fig. 4.1: Module architecture diagram.
- Fig. 4.2: Example Qiskit circuit for $n = 3$ MoS measurement.
- Fig. 4.3: Empirical vs. theoretical $\Pr[s \mid b = 1]$ validation plot.

Chapter 5 — Experiments

- Table 5.1: Theory-vs-empirics verdict for each tested prediction.
- Fig. 5.1: Noise sweep—$(1 - 2\eta)^2$ attenuation and non-monotonic acceptance.
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- Fig. 5.5: Verifier truncation—acceptance vs. samples at multiple ε.
- Table 5.2: Dishonest prover rejection rates.
- Table 5.3: k-sparse misclassification vs. theoretical bound.

5 Alignment with Marking Criteria

5.1 Technical Content

Use of references. Chapter 2 surveys Mahadev (2018), Caro et al. (ITCS 2024, arXiv 2024), Ma–Su–Deng (2024), Goldwasser–Kalai et al., and the impossibility landscape. Goes beyond immediately relevant prior work to position the project in the broader verification-of-computation and verification-of-learning context (per Dr. Caro's feedback).

Problem solving and tools. End-to-end implementation of a 2024 research paper's protocol. 11 experiments systematically isolating failure modes across the protocol's parameter space. Clean explanatory models for non-obvious findings (gate noise crossover, truncation inversion, small-n anomalies).

Effective problem analysis. Each experiment targets a specific theoretical prediction and reports a precise verdict (confirmed, partially confirmed, novel). The theory-vs-empirics table (§5.6) provides a single-page summary of all findings.

Technical achievement. Faithful simulation of a 2024 verification protocol with 11 experiments covering completeness, soundness, robustness, and sensitivity. Novel gate noise characterisation with an explanatory model not present in the literature. Precise identification of where theoretical preconditions are violated and how this manifests empirically.

Critical discussion. Chapter 6 ($\sim$1,100 words) critically evaluates: where bounds are tight, where the protocol breaks, gate noise as an open frontier, and limitations with explicit assessment of how each affects the conclusions. The Mahadev connection is discussed precisely (which step breaks, why), not hand-waved.

5.2 Communication Skills

Structure. Clear arc: motivation $\rightarrow$ background (three pillars unified) $\rightarrow$ protocol $\rightarrow$ implementation $\rightarrow$ experiments (grouped by property) $\rightarrow$ discussion $\rightarrow$ conclusions. Forward pointers between background subsections. Connecting diagram (Fig. 2.1).

Presentation. 8 figures and 3+ tables break up dense material. Pseudocode communicates protocol structure without code dumps. $\sim$8,500 words within the required range.

6 Honest Assessment: Risks and Mitigations

Innovative design—moderate. The project is a faithful simulation, not a new algorithm or protocol. The original empirical contributions are: the gate noise characterisation (novel finding with explanatory model), the systematic theory-vs-empirics comparison across 11 experiments, and the identification of weight-check dominance as a structural observation. An assessor looking for design novelty may still find this closer to "thorough empirical study." *Mitigation:* frame the gate noise result and the cross-cutting structural observations as genuine findings requiring non-trivial analysis, not just parameter sweeps.

No GL extraction. The original plan centred on scaling to $n = 16+$ via Goldreich–Levin. The actual implementation uses direct thresholding (Corollary 5). For single-parity targets, $|L| = 1$ throughout, so the extraction is never stressed. *Mitigation:* the k-sparse and bent-function experiments do stress the extraction at $|L| > 1$. GL is positioned as future work.

Discussion depth. The most critical risk from the original plan (superficial discussion) is now mitigated by having 11 experiments' worth of concrete findings to discuss. The risk shifts to *length*: fitting meaningful discussion of all findings into $\sim$1,100 words requires disciplined selection. *Mitigation:* the theory-vs-empirics table absorbs factual summaries; the prose focuses on interpretation, not repetition.

Background balance. Still a risk if the background reads as textbook exposition rather than targeted preparation. *Mitigation:* the restructured flow with forward pointers ensures each subsection serves a later chapter. Strict 1,600-word budget enforced.

Appendix: Prose Quality Guide

The progress report received specific criticism on writing quality: "informal," "stream-of-thought," "incomplete or imprecise sentences," "concepts introduced without sufficient connective explanation," and "terms undefined." This appendix provides concrete before/after examples drawn from the progress report to define the target quality for each recurring issue. Each example is labelled with the feedback point it addresses.

A.1 Disconnected concepts → forward pointers

Feedback (both assessors): "Multiple concepts are introduced without sufficient connective explanation." "Subsections that seem completely disjoint from each other."

Progress report (end of §2.1):

"Learning with errors. Let $\mathbb{Z}_q$ denote the ring of integers modulo q ... We say that an algorithm for this is efficient if, given samples, it outputs $y = f(x)$ with probability exponentially close to 1, in polynomial time in n."

Then §2.2 opens:

"We can define a trapdoor claw-free function family as follows: ..."

LWE ends; TCF begins with no connection between them. A reader unfamiliar with the field sees two unrelated definitions.

Improved:

"... in polynomial time in n.
The hardness of the LWE problem is not merely of independent interest: it is the computational assumption underlying the cryptographic primitives used in quantum verification. Specifically, Mahadev's protocol [**?**] relies on *trapdoor claw-free* (TCF) function families, whose security can be based on LWE [**?**]. We now define these."

Rule: every subsection boundary must contain 1–2 sentences explaining *why* the next topic follows from the previous one and *where* it will be used later. For the final report's restructured Chapter 2, every subsection should end with an italic *Forward pointer* sentence (as specified in the plan).

A.2 Prose definitions → formal notation

Feedback (first assessor): "Some definitions would be clearer if expressed using mathematical notation rather than long sentences."

Progress report (§2.4):

"The goal in both types of learning is to learn an almost-optimal approximating function compared to a benchmark class $\mathcal{B}$, such that given accuracy parameter ε, confidence parameter δ, and access to training data i.i.d. from D, an α-agnostic learner has to output, with success probability $\geq 1 - \delta$ a hypothesis h such that: [equation]"

"In simpler words, that the hypothesis function outputs 'incorrect answers' at most as likely as the best benchmark (by metric of most likely to be accurate) for any input x, multiplied by α with some additive error margin ε."

The definition is buried in a run-on sentence, then restated informally in a way that is arguably less precise than the equation itself.

Improved:

Definition (α-agnostic learner). Let D be a distribution over $\{0,1\}^n \times \{0,1\}$ and let $\mathcal{B}$ be a benchmark concept class. An algorithm $\mathcal{A}$ is an *α-agnostic learner* for $\mathcal{B}$ with accuracy ε and confidence δ if, given i.i.d. samples from D, it outputs a hypothesis h satisfying

$$\Pr_{(x,y)\sim D}[h(x) \neq y] \;\leq\; \alpha \cdot \inf_{b\in\mathcal{B}} \Pr_{(x,y)\sim D}[b(x) \neq y] \;+\; \varepsilon$$

with probability at least $1-\delta$. The first term captures how far the best benchmark function is from perfect prediction; the learner is allowed to incur α times that irreducible error, plus an additive tolerance ε.

Rule: use `\begin{definition}` environments (from `amsthm`) for all formal notions: α-agnostic learner, promise class $\mathcal{D}^{\text{func}}_{U_n;\geq\theta}$, completeness/soundness (Definition 7), QPIP model. Follow each definition with at most one sentence of plain-language gloss. Never restate the same definition twice in different words.

A.3 Vague problem statement → precise research question

Feedback (second assessor): "Section 1.2 should contain the actual project goal, but it is written in an obscure style."

Progress report (§1.2):

"The central challenge is therefore to make targeted optimizations on existing verification schemes, surgically reducing their demands on critical resources from client memory, communication bandwidth, to coherence times. I aim to do this whilst building in robustness against noise and decoherence effects, bridging an implementation gap and modifying a protocol to operate effectively within realistic constraints to find a concrete path toward making secure, delegated QML a practical reality on the hardware of the near future."

This is 70 words of aspiration with no concrete deliverable. It does not say *which* protocol, *which* resource, or *what counts as success*.

Improved (§1.2 of the final report):

"This project asks: *can a classical client efficiently verify that an untrusted quantum server has correctly performed agnostic parity learning?* We investigate this question by implementing the mixture-of-superpositions (MoS) verification protocol of Caro et al. [**?**] and empirically characterising its completeness, soundness, and robustness across 11 experiments covering system sizes $n = 4$ to 16, adversarial provers, label-flip and gate-level noise, Fourier-dense targets, and verifier sample budgets."

Rule: the research question should be a single italicised sentence. Follow it immediately with a concrete statement of *how* the project answers it and *what* the deliverables are.

A.4 Objectives not referenced → explicit traceability

Feedback (second assessor): "The student had set specific objectives in the project specification, but these objectives are not referred to in the project report, hence it is difficult to measure the progress that is achieved."

Progress report:

Objectives listed in the specification (§2.1.1–2.2.1) are never mentioned again in the progress report body. Chapter 3 describes implementation work without connecting it to any specific objective.

Improved (example from §5.4 of the final report):

"**Experiment 3.2: Gate-level depolarising noise.** This experiment addresses **O3** (empirical confirmation of theoretical predictions) and **O4** (novel gate noise characterisation). No existing theorem in [**?**] covers circuit-level noise; this is the first empirical evidence . . . "

And in the Discussion (§6.2):

"Objective O1 (faithful implementation) is fully achieved: the distributional protocol (Theorems 12 and 15) is implemented end-to-end with 97+ unit tests. Objective O4 (novel findings) is achieved via the gate noise characterisation (§5.4, Exp 3.2), which identifies a sharp phase transition . . . "

Rule: number the objectives **O1**–**O5** in §1.3. Reference these labels when introducing each experiment and when evaluating results in the Discussion.

A.5 Informal / stream-of-thought → precise, concise

Feedback (first assessor): "The language is at times informal . . . parts of the report read more like a stream-of-thought than a carefully constructed exposition."

Progress report (§2, opening):

"A large section of work completed during this project was reading into the necessary materials within the field to find enough direction to follow through with the specification. The initial scope of the project required exploration into two key papers [Mah23] [Car+24]. These ideas from Mahadev and Caro et al. will be foundation of what we are building on, hopefully finding a way to bring core themes from the two papers together."

"Large section of work" is vague, "reading into" is informal, "hopefully finding a way" signals uncertainty where the chapter should project authority.

Improved:

"This chapter introduces the three theoretical foundations of the verification protocol studied in this project: quantum computation (§2.1), the verification of quantum computation (§2.2), and agnostic learning via Boolean Fourier analysis (§2.3–2.4). These threads converge in §2.5, which presents the MoS-based verification framework of Caro et al. [**?**] that the remainder of the report implements and evaluates."

Rule: chapter openings should be a roadmap in 2–3 sentences. State what the chapter covers, in what order, and why. Do not narrate the research process ("I read. . . ", "we explored. . . "); present the content directly.

A.6 Ambiguous citation → distinct keys

Feedback (second assessor): "The reference 'Caro et al.' should be given properly . . . This is particularly important, because there is [Car+24] and [Car+25]."

Progress report (§3.1):

". . . we identified that the most important focus for implementation so far would be to implement the MoS model from Caro et al. [Car+24]."

Later: "Caro et al.'s paper [Car+24] in Definition 8 states: . . . "

And in §3.2: "A natural point of contact between Mahadev-style classical verification and the MoS concept of Caro et al. is . . . " [no citation at all].

Improved:

Use distinct keys throughout: `[?]| for the ITCS 2024 paper and [?]| for the arXiv impossibility paper. Every mention of either paper gets the appropriate citation. Never write ``Caro et al.'' without a [?]|. Never write ``the paper'' when two Caro et al. papers are in scope---use the short descriptor (``the verification protocol [?]'' vs. ``the impossibility result [?]'').`

A.7 Experiment results disconnected from theory → claim-first structure

Feedback (Dr. Caro, plan meeting): "When discussing your results, connect back to the research question and the objectives." *Assessor feedback:* "Implementation not discussed in a way that could be verified by a project outsider."

Hypothetical bad pattern:

"We ran the noise sweep experiment with η from 0 to 0.4. Figure 5.1 shows the results. The acceptance rate drops and then increases. The weight tracks $(1-2\eta)^2$. This is consistent with Lemma 6."

This presents results in the order they were *observed*, not the order a reader needs. The theoretical claim comes last, as an afterthought.

Improved (claim-first structure):

"**Experiment 3.1** tests the MoS noise model (Definition 5(iii)) and the resulting Fourier weight attenuation predicted by Lemma 6: under label-flip noise at rate η, the accumulated Fourier weight should scale as $(1-2\eta)^2$.
Result. The prediction is precisely confirmed. Observed median weight tracks $(1-2\eta)^2$ with less than 3% relative error across all n and η (Figure 5.1a). Acceptance remains above 69% for all tested parameters; the theoretical breakdown at $\eta_{\max} = 0.447$ lies beyond the tested range.
Non-obvious finding. Acceptance is non-monotonic in η: it dips at moderate noise ($\eta = 0.05$–0.15) then recovers at high noise (Figure 5.1b). The mechanism is that the *relative* margin ($\mathrm{eps}^2/(8a^2)$) grows as a^2 shrinks, making the weight check proportionally more forgiving even as absolute signal weakens."

Rule: every experiment in Chapter 5 should follow this template: (1) State the theoretical claim being tested, with a citation. (2) State the result—confirmed, partially confirmed, or novel—in bold. (3) Present the evidence (figure/table reference, key numbers). (4) Note any non-obvious findings or caveats.